

Eigenstructure of **A** ~~work~~ in appended PDF. (using maple)

eigenvalues: $\lambda_1 = u - a$, $\lambda_2 = u + a$, $\lambda_3 = \lambda_4 = u$

right eigenvectors: $\vec{r}_1 = \begin{bmatrix} -P/a \\ 1 \\ 0 \\ -ap \end{bmatrix}$, $\vec{r}_2 = \begin{bmatrix} 1 \\ a/p \\ 0 \\ a^2 \end{bmatrix}$, $\vec{r}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{r}_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

left eigenvectors: $\vec{l}_1 = [0, \frac{1}{2}, 0, -\frac{1}{2ap}]$
 $\vec{l}_2 = [0, \frac{P}{2a}, 0, \frac{1}{2a^2}]$
 $\vec{l}_3 = [1, 0, 0, -\frac{1}{a^2}]$
 $\vec{l}_4 = [0, 1, 0, 0]$

Eigenstructure of **B**

eigenvalues: $\lambda_1 = v - a$, $\lambda_2 = v + a$, $\lambda_3 = \lambda_4 = v$

right eigenvectors: $\vec{r}_1 = \begin{bmatrix} -P/a \\ 0 \\ 1 \\ -ap \end{bmatrix}$, $\vec{r}_2 = \begin{bmatrix} 1 \\ 0 \\ a/p \\ a^2 \end{bmatrix}$, $\vec{r}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\vec{r}_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

left eigenvectors: $\vec{l}_1 = [0, 0, \frac{1}{2}, -\frac{1}{2ap}]$
 $\vec{l}_2 = [0, 0, \frac{P}{2a}, \frac{1}{2a^2}]$
 $\vec{l}_3 = [1, 0, 0, -\frac{1}{a^2}]$
 $\vec{l}_4 = [0, 1, 0, 0]$

Eigenstructure of

A

eigenvalues: $\lambda_1 = u-a$, $\lambda_2 = u+a$, $\lambda_3 = \lambda_4 = u$ (Same as **A**)

right eigenvectors:

$$\vec{r}_1 = \left[1, u-a, v, \frac{(-2au + u^2 + v^2)\gamma - u^2 + 2au - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_2 = \left[1, u+a, v, \frac{(2au + u^2 + v^2)\gamma - u^2 - 2au - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_3 = \left[1, u, 0, \frac{u^2}{2} - \frac{v^2}{2} \right]^T$$

$$\vec{r}_4 = \left[0, 0, 1, v \right]^T$$

left eigenvectors:

$$\vec{l}_1 = \left[\frac{u^2(\gamma-1) + 2au + v^2(\gamma-1)}{4a^2}, \frac{-\gamma u - a + u}{2a^2}, \frac{v(\gamma-1)}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_2 = \left[\frac{u^2(\gamma-1) - 2au + v^2(\gamma-1)}{4a^2}, \frac{-\gamma u + a + u}{2a^2}, \frac{v(\gamma-1)}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_3 = \left[\frac{2a^2 - (u^2 + v^2)(\gamma-1)}{2a^2}, \frac{u(\gamma-1)}{a^2}, \frac{v(\gamma-1)}{a^2}, \frac{-\gamma+1}{a^2} \right]$$

$$\vec{l}_4 = \left[-\frac{(u^2 + v^2)v(\gamma-1)}{2a^2}, \frac{uv(\gamma-1)}{a^2}, \frac{a^2 + v^2(\gamma-1)}{a^2}, -\frac{v(\gamma-1)}{a^2} \right]$$

Eigenstructure of

B

eigenvalues: $\lambda_1 = v-a$, $\lambda_2 = v+a$, $\lambda_3 = \lambda_4 = v$ (Same as **B**)

right eigenvectors:

$$\vec{r}_1 = \left[1, u, v-a, \frac{(-2av + u^2 + v^2)\gamma - u^2 + 2av - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_2 = \left[1, u, v+a, \frac{(2av + u^2 + v^2)\gamma - u^2 - 2av - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_3 = \left[1, 0, v, -\frac{u^2}{2} + \frac{v^2}{2} \right]^T$$

$$\vec{r}_4 = \left[0, 1, 0, u \right]^T$$

left eigenvectors:

$$\vec{l}_1 = \left[\frac{v^2(\gamma-1) + 2av + u^2(\gamma-1)}{4a^2}, \frac{u(\gamma-1)}{2a^2}, \frac{-\gamma v - a + v}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_2 = \left[\frac{v^2(\gamma-1) - 2av + u^2(\gamma-1)}{4a^2}, \frac{u(\gamma-1)}{2a^2}, \frac{-\gamma v + a + v}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_3 = \left[\frac{2a^2 - (u^2 + v^2)(\gamma-1)}{2a^2}, \frac{u(\gamma-1)}{a^2}, \frac{v(\gamma-1)}{a^2}, \frac{-\gamma+1}{a^2} \right]$$

$$\vec{l}_4 = \left[-\frac{(u^2 + v^2)u(\gamma-1)}{2a^2}, \frac{a^2 + u^2(\gamma-1)}{a^2}, \frac{uv(\gamma-1)}{a^2}, -\frac{u(\gamma-1)}{a^2} \right]$$

> **# Reset maple instance**

> restart;
> with(linalg):

> **# Set up input vectors representing 2-D inviscid Euler equations**

> W:=array(1..4, [rho, u, v, p]);
> U:=array(1..4, [rho, rho*u, rho*v, (p/(g-1)) + (rho * (u^2 + v^2))/2]);
> F:=array(1..4, [rho*u, rho*u^2+p, rho*u*v, rho*u*((g*p)/((g-1)*rho) + (u^2+v^2)/2)]);
> G:=array(1..4, [rho*v, rho*u*v, rho*v^2 + p, rho*v*((g*p)/((g-1)*rho) + (u^2+v^2)/2)]);

$$W := \begin{bmatrix} \rho & u & v & p \end{bmatrix}$$

$$U := \begin{bmatrix} \rho & \rho u & \rho v & \frac{p}{g-1} + \frac{\rho(u^2 + v^2)}{2} \end{bmatrix}$$

$$F := \begin{bmatrix} \rho u & \rho u^2 + p & \rho u v & \rho u \left(\frac{g p}{(g-1) \rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) \end{bmatrix}$$

$$G := \begin{bmatrix} \rho v & \rho u v & \rho v^2 + p & \rho v \left(\frac{g p}{(g-1) \rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) \end{bmatrix} \quad (1)$$

> **# Compute Jacobians and other matrix quantities**

> dUdW:=array(1..4,1..4):
> dUdW:=jacobian(U,W);

$$dUdW := \begin{bmatrix} 1 & 0 & 0 & 0 \\ u & \rho & 0 & 0 \\ v & 0 & \rho & 0 \\ \frac{u^2}{2} + \frac{v^2}{2} & \rho u & \rho v & \frac{1}{g-1} \end{bmatrix} \quad (2)$$

> dWdU:=array(1..4,1..4):
> dWdU:=inverse(dUdW);

(3)

$$dWdU := \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{u}{\rho} & \frac{1}{\rho} & 0 & 0 \\ -\frac{v}{\rho} & 0 & \frac{1}{\rho} & 0 \\ \frac{(u^2 + v^2)(g-1)}{2} & -u(g-1) & -v(g-1) & g-1 \end{bmatrix} \quad (3)$$

```
> dFdW:=array(1..4,1..4):
dFdW:=jacobian(F,W);
```

$$dFdW := \begin{bmatrix} u, \rho, 0, 0, \\ u^2, 2\rho u, 0, 1, \\ uv, \rho v, \rho u, 0, \\ u \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) - \frac{ugp}{\rho(g-1)}, \rho \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) + \rho u^2, \\ \rho uv, \frac{ug}{g-1} \end{bmatrix} \quad (4)$$

```
> dGdW:=array(1..4,1..4):
dGdW:=jacobian(G,W);
```

$$dGdW := \begin{bmatrix} v, 0, \rho, 0, \\ uv, \rho v, \rho u, 0, \\ v^2, 0, 2\rho v, 1, \\ v \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) - \frac{vgp}{\rho(g-1)}, \rho uv, \rho \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) \\ + \rho v^2, \frac{vg}{g-1} \end{bmatrix} \quad (5)$$

```
> dFdU:=array(1..4,1..4):
> dFdU:=evalm(dFdW&*dWdU):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do dFdU[i,j]:=
factor(simplify(dFdU[i,j])) od: od:
> evalm(dFdU);
```

(6)

$$\left[\begin{bmatrix} 0, 1, 0, 0 \end{bmatrix}, \right. \quad (6)$$

$$\left[-\frac{3}{2} u^2 + \frac{1}{2} u^2 g + \frac{1}{2} g v^2 - \frac{1}{2} v^2, -u(g-3), -v(g-1), g-1 \right],$$

$$\left[-uv, v, u, 0 \right],$$

$$\left[\frac{u(g^2 \rho u^2 + g^2 \rho v^2 - 3 \rho u^2 g - 3 \rho v^2 g + 2 \rho u^2 + 2 \rho v^2 - 2 g p)}{2 \rho (g-1)}, \right.$$

$$\left. -\frac{2 g^2 \rho u^2 - 5 \rho u^2 g - \rho v^2 g + 3 \rho u^2 + \rho v^2 - 2 g p}{2 \rho (g-1)}, -uv(g-1), u g \right]$$

> dGdU:=array(1..4,1..4):

> dGdU:=evalm(dGdW&*dWdU):

> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do dGdU[i,j]:=factor(simplify(dGdU[i,j])) od: od:

> evalm(dGdU);

$$\left[\begin{bmatrix} 0, 0, 1, 0 \end{bmatrix}, \right. \quad (7)$$

$$\left[-uv, v, u, 0 \right],$$

$$\left[-\frac{3}{2} v^2 + \frac{1}{2} u^2 g + \frac{1}{2} g v^2 - \frac{1}{2} u^2, -u(g-1), -v(g-3), g-1 \right],$$

$$\left[\frac{v(g^2 \rho u^2 + g^2 \rho v^2 - 3 \rho u^2 g - 3 \rho v^2 g + 2 \rho u^2 + 2 \rho v^2 - 2 g p)}{2 \rho (g-1)}, -uv(g \right.$$

$$\left. -1), -\frac{2 g^2 \rho v^2 - \rho u^2 g - 5 \rho v^2 g + \rho u^2 + 3 \rho v^2 - 2 g p}{2 \rho (g-1)}, v g \right]$$

> # Compute matrices A and B

> A:=array(1..4,1..4):

> A:=simplify(evalm(dWdU&*dFdU&*dUdW));

$$A := \begin{bmatrix} u & \rho & 0 & 0 \\ 0 & u & 0 & \frac{1}{\rho} \\ 0 & 0 & u & 0 \\ 0 & g p & 0 & u \end{bmatrix}$$

(8)

> B:=array(1..4,1..4):

> B:=simplify(evalm(dWdU&*dGdU&*dUdW));

$$B := \begin{bmatrix} v & 0 & \rho & 0 \\ 0 & v & 0 & 0 \\ 0 & 0 & v & \frac{1}{\rho} \\ 0 & 0 & g\rho & v \end{bmatrix} \quad (9)$$

> # Outline algebraic assumptions

```
> assume(g, real);
> assume(g > 0);
> assume(rho, real);
> assume(rho > 0);
> assume(a, real);
> assume(a > 0);
> assume(p, real);
> p:=rho*a^2/g;
```

$$p := \frac{\rho \sim a \sim^2}{g \sim} \quad (10)$$

> # Compute eigenvalues of A, script A, B and script B

```
>
# A and script A are similar matrices, so we can show that they have the same eigenvalues
```

```
I4:=array(1..4,1..4, identity):
```

```
> char_poly1A:=factor(simplify(det(A-lambda*I4)));
> eigenvals1A:=solve(char_poly1A=0, lambda);
> char_poly2A:=factor(simplify(det(dFdU-lambda*I4)));
> eigenvals2A:=solve(char_poly2A=0, lambda);
```

$$\text{eigenvals1A} := u - a \sim, u + a \sim, u, u$$

$$\text{eigenvals2A} := u - a \sim, u + a \sim, u, u$$

(11)

```
>
# The same is true of B and script B
```

```
> char_poly1B:=factor(simplify(det(B-lambda*I4)));
> eigenvals1B:=solve(char_poly1B=0, lambda);
> char_poly2B:=factor(simplify(det(dGdU-lambda*I4)));
> eigenvals2B:=solve(char_poly2B=0, lambda);
```

$$\text{eigenvals1B} := v - a \sim, v + a \sim, v, v$$

$$\text{eigenvals2B} := v - a \sim, v + a \sim, v, v$$

(12)

> # Find eigenvectors using the eigenvalues

```

> # initialize arrays to hold the eigenvectors
rightAeigenvectors:=array(1..4,1..4):
rightsAeigenvectors:=array(1..4,1..4):
rightBeigenvectors:=array(1..4,1..4):
rightsBeigenvectors:=array(1..4,1..4):

```

begin with the eigenvectors of A

```

>
# eigenvalue 1 of A (eigenvector 1)
AminLam1:=array(1..4,1..4):
> AminLam1:=evalm(A-evalm((eigenvals1A[1])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> r1A:= nullspace(AminLam1)[1];

```

$$r1A := \begin{bmatrix} -\frac{\rho}{a} & 1 & 0 & -a\rho \end{bmatrix} \quad (13)$$

```

>
# eigenvalue 2 of A (eigenvector 2)
AminLam1:=evalm(A-evalm((eigenvals1A[2])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> r2A:=nullspace(AminLam1)[1];

```

$$r2A := \begin{bmatrix} 1 & \frac{a}{\rho} & 0 & a^2 \end{bmatrix} \quad (14)$$

```

>
# eigenvalue 3 of A (eigenvectors 3 and 4 since this is a repeated eigenvalue)
AminLam1:=evalm(A-evalm((eigenvals1A[3])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> nullspaceA3:=nullspace(AminLam1):
> r3A:=nullspaceA3[1];

```

$$r3A := \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad (15)$$

```

> r4A:=nullspaceA3[2];

```

$$r4A := \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \quad (16)$$

next compute the eigenvectors of script A

```

>
# eigenvalue 1 of sA (eigenvector 1)
AminLam1:=array(1..4,1..4):
> AminLam1:=evalm(dFdU-evalm((eigenvals1A[1])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> r1sA:= simplify(nullspace(AminLam1)[1]);

```

$$r1sA := \begin{bmatrix} 1 & u - a \sim & v & \frac{(-2 a \sim u + u^2 + v^2) g \sim - u^2 + 2 a \sim u - v^2 + 2 a \sim^2}{2 g \sim - 2} \end{bmatrix} \quad (17)$$

```
>
# eigenvalue 2 of sA (eigenvector 2)
AminLaml:=evalm(dFdU-evalm((eigenvals1A[2])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, AminLaml[i,j])) od: od:
> r2sA:=simplify(nullspace(AminLaml)[1]);
```

$$r2sA := \begin{bmatrix} 1 & u + a \sim & v & \frac{(2 a \sim u + u^2 + v^2) g \sim - u^2 - 2 a \sim u - v^2 + 2 a \sim^2}{2 g \sim - 2} \end{bmatrix} \quad (18)$$

```
>
# eigenvalue 3 of sA (eigenvectors 3 and 4 since this is a repeated eigenvalue)
AminLaml:=evalm(dFdU-evalm((eigenvals1A[3])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, AminLaml[i,j])) od: od:
> nullspaceA3:=nullspace(AminLaml):
> r3sA:=simplify(nullspaceA3[1]);
```

$$r3sA := \begin{bmatrix} 1 & u & 0 & \frac{u^2}{2} - \frac{v^2}{2} \end{bmatrix} \quad (19)$$

```
> r4sA:=simplify(nullspaceA3[2]);
```

$$r4sA := \begin{bmatrix} 0 & 0 & 1 & v \end{bmatrix} \quad (20)$$

```
>
# eigenvalue 4 of sA (should have an identical output of eigenvalue 3)
AminLaml:=evalm(dFdU-evalm((eigenvals1A[4])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, AminLaml[i,j])) od: od:
> nullspace(AminLaml);
```

$$\left\{ \begin{bmatrix} 0 & 0 & 1 & v \end{bmatrix}, \begin{bmatrix} 1 & u & 0 & \frac{u^2}{2} - \frac{v^2}{2} \end{bmatrix} \right\} \quad (21)$$

```
>
# next compute the eigenvectors of B
```

```
>
# eigenvalue 1 of B (eigenvector 1)
BminLaml:=array(1..4,1..4):
> BminLaml:=evalm(B-evalm((eigenvals1B[1])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, BminLaml[i,j])) od: od:
> r1B:=nullspace(BminLaml)[1];
```

$$r1B := \begin{bmatrix} -\frac{\rho \sim}{a \sim} & 0 & 1 & -a \sim \rho \sim \end{bmatrix} \quad (22)$$

eigenvalue 2 of B (eigenvector 2)

BminLaml:=evalm(B-evalm((eigenvals1B[2])*I4)):

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, BminLaml[i,j])) od: od:
> r2B:=nullspace(BminLaml)[1];
```

$$r2B := \begin{bmatrix} 1 & 0 & \frac{a\sim}{\rho\sim} & a\sim^2 \end{bmatrix} \quad (23)$$

>

eigenvalue 3 of B (eigenvectors 3 and 4 since this is a repeated eigenvalue)

BminLaml:=evalm(B-evalm((eigenvals1B[3])*I4)):

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, BminLaml[i,j])) od: od:
> r3B:=nullspace(BminLaml)[1];
```

$$r3B := \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad (24)$$

```
> r4B:=nullspace(BminLaml)[2];
```

$$r4B := \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \quad (25)$$

>

eigenvalue 4 of B (should have an identical output of eigenvalue 3)

BminLaml:=evalm(B-evalm((eigenvals1B[4])*I4)):

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, BminLaml[i,j])) od: od:
> nullspace(BminLaml);
```

$$\left\{ \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \right\} \quad (26)$$

> # next compute the eigenvectors of script B

>

eigenvalue 1 of B (eigenvector 1)

BminLaml:=array(1..4,1..4):

```
> BminLaml:=evalm(dGdU-evalm((eigenvals1B[1])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, BminLaml[i,j])) od: od:
> r1sB:=simplify(nullspace(BminLaml)[1]);
```

$$r1sB := \begin{bmatrix} 1 & u & v - a\sim & \frac{(-2 a\sim v + u^2 + v^2) g\sim - u^2 - v^2 + 2 a\sim v + 2 a\sim^2}{2 g\sim - 2} \end{bmatrix} \quad (27)$$

>

eigenvalue 2 of B (eigenvector 2)

BminLaml:=evalm(dGdU-evalm((eigenvals1B[2])*I4)):

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLaml[i,j]
:=simplify(subs(p=rho*a^2/g, BminLaml[i,j])) od: od:
> r2sB:=simplify(nullspace(BminLaml)[1]);
```

$$r2sB := \begin{bmatrix} 1 & u & v + a\sim & \frac{(2 a\sim v + u^2 + v^2) g\sim - u^2 - v^2 - 2 a\sim v + 2 a\sim^2}{2 g\sim - 2} \end{bmatrix} \quad (28)$$

eigenvalue 3 of B (eigenvectors 3 and 4 since this is a repeated eigenvalue)

BminLamI:=evalm(dGdU-evalm((eigenvals1B[3])*I4)):

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLamI[i,j]
:=simplify(subs(p=rho*a^2/g, BminLamI[i,j])) od: od:
> r3sB:=simplify(nullspace(BminLamI)[1]);
```

$$r3sB := \begin{bmatrix} 1 & 0 & v & -\frac{u^2}{2} + \frac{v^2}{2} \end{bmatrix} \quad (29)$$

```
> r4sB:=simplify(nullspace(BminLamI)[2]);
```

$$r4sB := \begin{bmatrix} 0 & 1 & 0 & u \end{bmatrix} \quad (30)$$

```
>
```

eigenvalue 4 of B (should have an identical output of eigenvalue 3)

BminLamI:=evalm(dGdU-evalm((eigenvals1B[4])*I4)):

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLamI[i,j]
:=simplify(subs(p=rho*a^2/g, BminLamI[i,j])) od: od:
> nullspace(BminLamI);
```

$$\left\{ \begin{bmatrix} 0 & 1 & 0 & u \end{bmatrix}, \begin{bmatrix} 1 & 0 & v & -\frac{u^2}{2} + \frac{v^2}{2} \end{bmatrix} \right\} \quad (31)$$

```
>
```

Check the right eigenvalue problem residuals

```
> # check residuals for eigenvalue eigenvector pairs of matrix A
```

res:=array(1..4):

```
> res0:=evalm((A&*r1A)-eigenvals1A[1]*r1A):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]
)) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (32)$$

```
> res0:=evalm((A&*r2A)-eigenvals1A[2]*r2A):
```

```
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]
)) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (33)$$

```
> res0:=evalm((A&*r3A)-eigenvals1A[3]*r3A):
```

```
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]
)) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (34)$$

```
> res0:=evalm((A&*r4A)-eigenvals1A[4]*r4A):
```

```
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]
)) od:
```

```
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (35)$$

```
> # check residuals for eigenvalue eigenvector pairs of matrix script A
```

```
res:=array(1..4):
> res0:=evalm((dFdU&*r1sA)-eigenvals1A[1]*r1sA):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (36)$$

```
> res0:=evalm((dFdU&*r2sA)-eigenvals1A[2]*r2sA):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (37)$$

```
> res0:=evalm((dFdU&*r3sA)-eigenvals1A[3]*r3sA):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (38)$$

```
> res0:=evalm((dFdU&*r4sA)-eigenvals1A[4]*r4sA):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (39)$$

```
>
# check residuals for eigenvalue eigenvector pairs in matrix B
```

```
> res0:=evalm((B&*r1B)-eigenvals1B[1]*r1B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (40)$$

```
> res0:=evalm((B&*r2B)-eigenvals1B[2]*r2B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (41)$$

```
> res0:=evalm((B&*r3B)-eigenvals1B[3]*r3B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]))
```

```

)) od:
> evalm(res);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (42)$$


```

```

> res0:=evalm((B&*r4B)-eigenvals1B[4]*r4B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
> evalm(res);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (43)$$


```

```

>
# check residuals for eigenvalue eigenvector pairs in matrix script B

```

```

> res0:=evalm((dGdU&*r1sB)-eigenvals1B[1]*r1sB):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
> evalm(res);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (44)$$


```

```

> res0:=evalm((dGdU&*r2sB)-eigenvals1B[2]*r2sB):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
> evalm(res);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (45)$$


```

```

> res0:=evalm((dGdU&*r3sB)-eigenvals1B[3]*r3sB):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
> evalm(res);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (46)$$


```

```

> res0:=evalm((dGdU&*r4sB)-eigenvals1B[4]*r4sB):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
> evalm(res);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (47)$$


```

```

>
# Compute the eigenvectors for the left
eigenproblem

```

```

> # first compute for the non-script matrices

```

```

# make right eigenvector matrix for A

```

```

RA:=array(1..4,1..4, sparse):

```

```

> for i from 1 to 4 by 1 do RA[i,1]:=simplify(r1A[i]) od:
> for i from 1 to 4 by 1 do RA[i,2]:=simplify(r2A[i]) od:
> for i from 1 to 4 by 1 do RA[i,3]:=simplify(r3A[i]) od:
> for i from 1 to 4 by 1 do RA[i,4]:=simplify(r4A[i]) od:
=
>
  # make the left eigenvector matrix for A by inverting the right eigenvector
  matrix
> LA:=array(1..4,1..4, sparse):
> LA:=inverse(RA):
=

```

```

> l1A,l2A,l3A,l4A:= vector(4),vector(4),vector(4),vector(4):
> for i from 1 to 4 by 1 do l1A[i]:=simplify(LA[1,i]) od:
> for i from 1 to 4 by 1 do l2A[i]:=simplify(LA[2,i]) od:
> for i from 1 to 4 by 1 do l3A[i]:=simplify(LA[3,i]) od:
> for i from 1 to 4 by 1 do l4A[i]:=simplify(LA[4,i]) od:
>
> evalm(l1A);
> evalm(l2A);
> evalm(l3A);
> evalm(l4A);

```

$$\begin{bmatrix} 0 & \frac{1}{2} & 0 & -\frac{1}{2a\tilde{\rho}} \end{bmatrix} \\
 \begin{bmatrix} 0 & \frac{\rho}{2a} & 0 & \frac{1}{2a^2} \end{bmatrix} \\
 \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{a^2} \end{bmatrix} \\
 \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$$

(48)

```

> # make right eigenvector matrix for B
=

```

```

  RB:=array(1..4,1..4, sparse):
> for i from 1 to 4 by 1 do RB[i,1]:=simplify(r1B[i]) od:
> for i from 1 to 4 by 1 do RB[i,2]:=simplify(r2B[i]) od:
> for i from 1 to 4 by 1 do RB[i,3]:=simplify(r3B[i]) od:
> for i from 1 to 4 by 1 do RB[i,4]:=simplify(r4B[i]) od:
=
>

```

```

> # make the left eigenvector matrix for B by inverting the right eigenvector
  matrix
=

```

```

> LB:=array(1..4,1..4, sparse):
> LB:=inverse(RB):
=

```

```

> l1B,l2B,l3B,l4B:= vector(4),vector(4),vector(4),vector(4):
> for i from 1 to 4 by 1 do l1B[i]:=simplify(LB[1,i]) od:
> for i from 1 to 4 by 1 do l2B[i]:=simplify(LB[2,i]) od:

```

```

> for i from 1 to 4 by 1 do l3B[i]:=simplify(LB[3,i]) od:
> for i from 1 to 4 by 1 do l4B[i]:=simplify(LB[4,i]) od:
>
> evalm(l1B);
> evalm(l2B);
> evalm(l3B);
> evalm(l4B);

```

$$\begin{bmatrix} 0 & 0 & \frac{1}{2} & -\frac{1}{2 a \sim \rho \sim} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & \frac{\rho \sim}{2 a \sim} & \frac{1}{2 a \sim^2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{1}{a \sim^2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}$$

(49)

```

>
# Now use the primitive left eigenvector matrices and coordinate transform back
into the conserved eigenvectors

```

```

> # make right eigenvector matrix for A

```

```

RsA:=array(1..4,1..4, sparse):

```

```

> for i from 1 to 4 by 1 do RsA[i,1]:=simplify(r1sA[i]) od:
> for i from 1 to 4 by 1 do RsA[i,2]:=simplify(r2sA[i]) od:
> for i from 1 to 4 by 1 do RsA[i,3]:=simplify(r3sA[i]) od:
> for i from 1 to 4 by 1 do RsA[i,4]:=simplify(r4sA[i]) od:

```

```

>
# make the left eigenvector matrix for A by inverting the right eigenvector
matrix

```

```

> LsA:=array(1..4,1..4, sparse):
> LsA:=inverse(RsA):

```

```

> l1sA,l2sA,l3sA,l4sA:= vector(4),vector(4),vector(4),vector(4):
> for i from 1 to 4 by 1 do l1sA[i]:=simplify(LsA[1,i]) od:
> for i from 1 to 4 by 1 do l2sA[i]:=simplify(LsA[2,i]) od:
> for i from 1 to 4 by 1 do l3sA[i]:=simplify(LsA[3,i]) od:
> for i from 1 to 4 by 1 do l4sA[i]:=simplify(LsA[4,i]) od:

```

```

> evalm(l1sA);
> evalm(l2sA);
> evalm(l3sA);
> evalm(l4sA);

```

$$\left[\frac{u^2 (g \sim - 1) + 2 a \sim u + v^2 (g \sim - 1)}{4 a \sim^2}, \frac{-g \sim u - a \sim + u}{2 a \sim^2}, -\frac{v (g \sim - 1)}{2 a \sim^2}, \right.$$

$$\begin{aligned}
& \left[\frac{g-1}{2a^2} \right] \\
& \left[\frac{u^2(g-1) - 2au + v^2(g-1)}{4a^2}, \frac{-gu + a + u}{2a^2}, -\frac{v(g-1)}{2a^2}, \right. \\
& \left. \frac{g-1}{2a^2} \right] \\
& \left[\frac{2a^2 - (u^2 + v^2)(g-1)}{2a^2}, \frac{u(g-1)}{a^2}, \frac{v(g-1)}{a^2}, \frac{-g+1}{a^2} \right] \\
& \left[-\frac{(u^2 + v^2)v(g-1)}{2a^2}, \frac{uv(g-1)}{a^2}, \frac{a^2 + v^2(g-1)}{a^2}, -\frac{v(g-1)}{a^2} \right]
\end{aligned} \tag{50}$$

> # make right eigenvector matrix for B

RsB:=array(1..4,1..4, sparse):

> for i from 1 to 4 by 1 do RsB[i,1]:=simplify(r1sB[i]) od:

> for i from 1 to 4 by 1 do RsB[i,2]:=simplify(r2sB[i]) od:

> for i from 1 to 4 by 1 do RsB[i,3]:=simplify(r3sB[i]) od:

> for i from 1 to 4 by 1 do RsB[i,4]:=simplify(r4sB[i]) od:

>

> # make the left eigenvector matrix for B by inverting the right eigenvector matrix

LsB:=array(1..4,1..4, sparse):

LsB:=inverse(RsB):

> l1sB,l2sB,l3sB,l4sB:= vector(4),vector(4),vector(4),vector(4):

> for i from 1 to 4 by 1 do l1sB[i]:=simplify(LsB[1,i]) od:

> for i from 1 to 4 by 1 do l2sB[i]:=simplify(LsB[2,i]) od:

> for i from 1 to 4 by 1 do l3sB[i]:=simplify(LsB[3,i]) od:

> for i from 1 to 4 by 1 do l4sB[i]:=simplify(LsB[4,i]) od:

>

> evalm(l1sB);

> evalm(l2sB);

> evalm(l3sB);

> evalm(l4sB);

$$\begin{aligned}
& \left[\frac{v^2(g-1) + 2av + u^2(g-1)}{4a^2}, -\frac{u(g-1)}{2a^2}, \frac{-gv - a + v}{2a^2}, \frac{g-1}{2a^2} \right] \\
& \left[\frac{v^2(g-1) - 2av + u^2(g-1)}{4a^2}, -\frac{u(g-1)}{2a^2}, \frac{-gv + a + v}{2a^2}, \frac{g-1}{2a^2} \right] \\
& \left[\frac{2a^2 - (u^2 + v^2)(g-1)}{2a^2}, \frac{u(g-1)}{a^2}, \frac{v(g-1)}{a^2}, \frac{-g+1}{a^2} \right]
\end{aligned}$$

$$\left[\begin{array}{cccc} -\frac{(u^2 + v^2) u (g_{\sim} - 1)}{2 a_{\sim}^2} & \frac{a_{\sim}^2 + u^2 (g_{\sim} - 1)}{a_{\sim}^2} & \frac{u v (g_{\sim} - 1)}{a_{\sim}^2} & -\frac{u (g_{\sim} - 1)}{a_{\sim}^2} \end{array} \right] \quad (51)$$

Check the left eigenvalue problem residuals

> # begin by making the eigenvalue matrix of A and script A

```
eigenval_matrixA:=Matrix([[eigenvals1A[1],0,0,0],[0,eigenvals1A[2],0,0],[0,0,eigenvals1A[3],0],[0,0,0,eigenvals1A[4]]]);
```

$$eigenval_matrixA := \begin{bmatrix} u - a_{\sim} & 0 & 0 & 0 \\ 0 & u + a_{\sim} & 0 & 0 \\ 0 & 0 & u & 0 \\ 0 & 0 & 0 & u \end{bmatrix} \quad (52)$$

> # check left A eigenvector matrix

```
simplify(evalm(LA&*A - eigenval_matrixA&*LA));
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (53)$$

> # check left script A eigenvector matrix

```
simplify(evalm(LsA&*dFdU - eigenval_matrixA&*LsA));
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (54)$$

> # make eigenvalue matrix for B and script B

```
eigenval_matrixB:=Matrix([[eigenvals1B[1],0,0,0],[0,eigenvals1B[2],0,0],[0,0,eigenvals1B[3],0],[0,0,0,eigenvals1B[4]]]);
```

(55)

$$eigenval_matrixB := \begin{bmatrix} v - a_{\sim} & 0 & 0 & 0 \\ 0 & v + a_{\sim} & 0 & 0 \\ 0 & 0 & v & 0 \\ 0 & 0 & 0 & v \end{bmatrix} \quad (55)$$

>

check left B eigenvector matrix

simplify(evalm(LB*B - eigenval_matrixB*LB));

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(56)

>

check left script B eigenvector matrix

simplify(evalm(LsB*dGdU - eigenval_matrixB*LsB));

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(57)

>